Working as a Physicist: Approximation, Symmetry and Units: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10

Solution 1 — Dimensional fingerprint. Return →

Write T = C,m^a k^b with C dimensionless. Dimensions give {T} = {M}^a({M},{T}^(-2))^b = {M}^(a+b),{T}^(-2b). Matching powers: a+b=0 and -2b=1, so b=-(1)/(2), a=(1)/(2). Hence

$$T = C\sqrt{(m)/(k)}. \tag{1}$$

Dimensional analysis cannot fix C; the true value is 2π, which requires the equation of motion. **Check:** stiffer spring (larger k) gives a shorter period; heavier mass gives a longer one, both physically sensible.

Solution 2 — How big is a star? Return →

Check the dimensions of σ T^4: it is power per unit area, [{M},{T}^(-3)]. Multiplying by R^2 (an area) gives power [{M},{L}²,{T}^(-3)], the dimensions of luminosity. The relation is consistent.

Doubling R multiplies L by $2^2 = 4$. Doubling T multiplies L by 2^4=16. Temperature is the stronger lever because a hotter surface radiates more per unit area **and** the spectrum shifts to shorter, more energetic wavelengths. **Check:** a star twice as hot and the same size is 16× brighter in this model, which is why hot stars dominate the light and cool stars dominate the numbers.

Solution 3 — The mass of a raindrop's argument. Return →

(a) Let m = Cγ^a r^b g^c. Since [γ]={M},{T}^(-2), [r]={L}, [g]={L},{T}^(-2):

$$\{M\} = \{M\}^a,\{L\}^(b+c),\{T\}^(-2a-2c). \quad (2)$$

Matching: a=1; b+c=0; -2-2c=0 c=-1, b=1. So m ≈ γ r/g times a number of order one. (The exact combination is 2πγ r/g when the drop's weight matches the rim force.)

(b) Take m ≈ 2π γ r / g with g=9.8: at r=1 {mm}, m ≈ 2π(0.07)(10^(-3))/9.8 ≈ 4.5×10^(-5) {kg}. This is about ten times larger than a real 1 mm drop, so real drops detach at roughly r 0.3–0.5 {mm} before the weight exceeds the rim force. The estimate gives the correct scalings m ∝ γ r, so doubling γ doubles the hanging mass and doubling r doubles the mass a rim of that size can hold — meaning the **shape** of the drop sets the limit, not the simple balance alone. **Check:** the result is a mass, is proportional to surface tension, and becomes zero when gravity is infinite — all physically required.

Solution 4 — Small-angle error budget. Return →

(a) sin θ = θ - θ³/6 +, so the fractional error of using θ is θ²/6. At θ = 0.30: θ²/6 = 0.09/6 = 0.015, i.e. about 1.5% (it **underestimates** by that fraction).

(b) Set θ²/6 = 0.01 θ = √(0.06) = 0.245 {rad} ≈ 14°. Numerically sin (0.245)/0.245 - 1 = -0.0099, so the estimate is accurate to a few per cent of itself. **Check:** the approximation is generous enough for olympiad work below about 15°, and you can now say so with a number instead of a feeling.

Solution 5 — Reading a log–log slope. Return →

(a) n=Δlog y/Δlog x = (2.90-1.30)/2 = 0.80.

(b) At log x = 0, log y = 1.30 so k=10^(1.30) = 20 with units {m,s^(-0.8)}. Fractional powers look strange but are legitimate for empirical laws.

(c) The slope uses logarithms, so a factor-of-two error in one reading shifts a log by only 0.30; the average line barely moves. A naive two-point calculation of y/x would be ruined by that same point because errors enter additively. **Check:** a log–log plot converts multiplicative errors into additive ones, which is precisely why it is robust.

Solution 6 — Symmetry in a lift. Return →

In a freely falling frame every object shares the acceleration g downward, so **relative** to the car nothing accelerates. The pendulum hangs limp: the effective gravity is zero, so its period T=2π√(L/g_({eff})) becomes infinite. The block does not compress its spring beyond its natural length, and the scale reads zero. All three behave as though gravity had been switched off — this is the symmetry between a falling frame and a gravity-free frame.

With a downward acceleration of 0.5g, the effective gravity in the car is g_({eff}) = g - 0.5g = 0.5g: the pendulum swings with a period √(2) times longer, the block compresses the spring by half its usual amount, and the scale reads half the weight. **Check:** the limiting case a→ g reproduces the free-fall answer, and a→0 reproduces a lift at rest.

Solution 7 — Fermi: number of stars in the observable sky. Return →

(a) Photos needed: 4×10^4/10 = 4×10³. Stars: 4×10³×10^4 = 4×10^7, i.e. about 10^8 stars to order of magnitude.

(b) The survey's 10^9 is about 25× larger. Real wide-field images overlap and repeated coverage increases depth; more importantly, a survey's plate limit changes how many stars a square degree contains **non-linearly**, and "stars" usually includes faint distant stars in crowded fields that a simple count under-represents. A factor of order ten is exactly the honesty range of a four-factor chain. **Check:** the chain has three factors each known to about a factor of two, so a factor √(2³)≈3 random scatter plus systematic choices fully explains the gap.

Solution 8 — Power-law scaling of a body. Return →

(a) Surface area A ∝ s², mass m∝ s³, so A/m ∝ s^(-1): larger animals lose less heat per unit mass.

(b) With heat content ∝ m and loss ∝ A, the cooling time is τ ∝ m/A ∝ s. A body twice as large in every dimension holds its temperature roughly twice as long.

(c) Advantage: larger bodies retain heat in cold environments. Disadvantage: the same surface must supply oxygen and food for a growing volume, so internal transport must become specialised (lungs, circulation). The model is crude because it ignores metabolism, insulation, shape change and internal convection — a real organism's metabolic rate is not simply proportional to volume. **Check:** the exponent comes from the ratio of two geometric powers, which is the robust part; the coefficient is biology.

Solution 9 — The limits test that catches wrong formulas. Return →

(a) (i) $v^2R/(2GM-v^2R)$ has $\{L\}^2,\{T\}^{(-2)}\cdot\{L\}/(\{L\}^3,\{T\}^{(-2)}) = \{L\}$: correct. (ii) $vR/\sqrt{(GM)}$ has $\{L\},\{T\}^{(-1)}\cdot\{L\}/(\{L\}^{(3/2)},\{T\}^{(-1)}) = \{L\}^{(1/2)}$: wrong — a square root of a length is not a height. (iii) $v^2/(2g)$ has $\{L\}$, correct as a small-v limit.

(b) As $v\to0$, (i) and (iii) give $h\to0$; (ii) gives zero too. So the small-v limit alone does not discriminate (i) from (iii).

(c) As $v^2R \to 2GM$ — the escape condition — (i) gives $h\to\infty$, which is exactly right: at escape speed the projectile never returns. Formula (iii) gives a finite height and is therefore wrong for speeds approaching escape. Formula (i) is the correct energy result and reduces to $v^2/2g$ when $v^2R \ll GM$. **Check:** only (i) is monotonic in v over the whole allowed range and diverges at escape.

Solution 10 — Building a model from scratch. Return →

(a) Let $R = C\,E^a\,\rho^b\,g^c$. Dimensions: $\{L\} = (\{M\},\{L\}^2,\{T\}^{(-2)})^a(\{M\},\{L\}^{(-3)})^b(\{L\},\{T\}^{(-2)})^c$. Mass: $a+b=0$. Time: $-2a-2c=0$ $c=-a$. Length: $2a-3b+c=1$ $2a+3a-a = 4a = 1$, so $a=(1)/(4)$, $b=-(1)/(4)$, $c=-(1)/(4)$:

$$R = C((E)/(\rho\, g))^{(1/4)}. \tag{3}$$

(b) A strength term Y (pressure, dimensions $[\{M\},\{L\}^{(-1)},\{T\}^{(-2)}]$) enters when the target is strong rather than gravity-dominated. Then a dimensionless ratio such as $E/(Y\,R^3)$ can appear and the single power law becomes a function of a dimensionless group — the crater transitions from strength- to gravity-dominated.

(c) At hypervelocity, the impactor is vaporised and the energy is delivered as a blast, so its shape is irrelevant — only E survives. At low speed the impactor simply presses a hole of its own size, so its radius a must enter. Every such regime change is a missing quantity, and dimensional analysis tells you **which** missing quantity is needed by where the fit fails.